SANKALP

Robotics Mechanical Design Engineer

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SUMMARY

Mechanical design engineer building end-effectors and actuation mechanisms for autonomous construction robots and cobots at Origin. Full development loop from concept CAD and FEA/CFD through DFMA, sheet-metal design and prototype testing. 2 published patent applications, 1 ASME paper, and national wins incl. **AIR 1, Autodesk Design Challenge** and **Smart India Hackathon 2024 (Hardware)**.

EXPERIENCE

Origin — Autonomous Construction Robotics **Bengaluru | May 2025 – Present**

Robotics Mechanical Design Engineer (Jul 2026 – Present) · Intern (May 2025 – Jul 2026), PPO Dec 2025

- Designed **swappable sander and sprayer end-effector systems** in Onshape, letting a single autonomous robot platform switch between surface-finishing tasks on construction sites.
- Designed an **offset sander end-effector for a UR12e cobot** with **servoing and vision capabilities** for robotic surface finishing.
- Engineered a **pneumatic-cylinder-actuated bucket-swap mechanism** for fast tool and material changeover on the robot.
- Re-designed structural and tooling parts for **sheet-metal fabrication using DFMA** to simplify manufacturing and assembly.
- Earned a **Pre-Placement Offer (PPO) in Dec 2025** (7th semester) for ownership of robot tooling sub-systems; joined full-time in Jul 2026.

Adelix Pvt. Ltd. — Design-for-Manufacturing Intern **Aug 2024 – Oct 2024**

- Designed a **universal docking system for medium-class multirotor UAVs**, including a gear- and drag-based cable uncoiling/recoiling mechanism driven by integrated motor drives.

Anusandhan National Research Foundation (DST, Govt. of India) — Summer Research Intern **Jun 2024**

- Designed a tabletop fume-hood ventilation system; optimised airflow via CFD and structure via topology optimisation.

PROJECTS

Autonomous Multirotor UAV [Report] | *SolidWorks, Ansys Fluent/Structural, LS-Dyna, ArduPilot, OpenCV* **May 2024 – Jul 2024**

- Designed a 2 kg hybrid-H quadcopter (602 mm wheelbase; Pixhawk + Jetson Nano) achieving **2.54:1 thrust-to-weight** (49.7 N) and **15 min endurance**.
- Validated the chassis via Ansys modal analysis (first natural frequency 252 Hz) and an LS-Dyna 30 m drop simulation (**FoS 2.36**); redesigned the stack cover to cut drag **21%** (2.08 → 1.65 N).

eVTOL Quad-plane for Medical Logistics [Report] | *Fusion 360, Autodesk CFD, ROS 2, Gazebo, URDF* **Mar 2024 – Dec 2024**

- Sized a hybrid VTOL for **40 kg payload** and 50+ min endurance with a morphing wing that reconfigures in transition to reduce cruise drag (CFD-validated); **patent published**.
- Generated a full-vehicle URDF (links, joints, inertials, transmissions) from CAD and ran controller-in-the-loop missions, payload drop and waypoint tests in **ROS 2 + Gazebo**.

Extrusion-Based Ceramic 3D Printer (B_4C) [Deck] | *SolidWorks, Ansys Fluent, Arduino/C++* **Feb 2024 – May 2024**

- Retrofitted a Creality Ender 3 with a syringe-pump extruder (NEMA 23, TB6600, lead screw) and **closed-loop Arduino control** with ultrasonic feedstock sensing and alerts; led to a **patent application and ASME paper**.
- Characterised 5 slurry compositions and validated nozzle flow in Ansys Fluent at 10% and 100% feed rates.

PATENTS & PUBLICATIONS [Patent filings]

- **Patent (published):** An Extrusion System for 3D Printing and a Method of Operation Thereof — IN App. 202521020630 (2025); Design 450682-001.
- **Patent (published):** An Electric (eVTOL) Aircraft for Medical Logistics and Disaster Surveillance — IN App. 202521057108 (2025); Design 462171-001.
- **Paper:** Evaluating Extrusion Dynamics in Ceramic 3D Printing Using FEA — ASME IMECE India 2025, IMECE-INDIA2025-161187.

AWARDS & CERTIFICATIONS [Certificates]

- **AIR 1**, Autodesk Design Challenge 2024 (IIT Kanpur) · **Winner**, Smart India Hackathon 2024 (Hardware) · **Best Autonomous UAV**, SAE Aerothon 2024 · **AIR 2**, Best CFD Analysis, SAE ISS Drone Development Challenge 2024
- Certified SolidWorks Associate (CSWA) · Top 5% nationwide, Aircraft Design (NPTEL, IIT Kanpur) [Portfolio]

SKILLS

CAD & Design	Onshape, SolidWorks (CSWA), Fusion 360; sheet-metal, DFMA, mechanism design, drawings
Simulation	Ansys Structural/Modal/Fluent, LS-Dyna, Autodesk CFD, XFLR5; FEA, CFD, topology optimisation
Robotics & Software	Python, C++, OpenCV, ROS 2, Gazebo, URDF, ArduPilot, Pixhawk, Jetson Nano, Arduino
Manufacturing	Additive manufacturing (FDM, ceramic extrusion), sheet-metal fabrication, prototyping

EDUCATION

B.Tech, Mechanical Engineering — IIITDM Jabalpur **2022 – 2026 | CPI 8.6/10**